Replication Report:
Many-Body Order Parameters for Multipoles in Solids

Kang, Shiozaki, Cho — arXiv:1812.06999 (Phys. Rev. B **100**, 245134, 2019)

Automated replication (OpenClaw subagent)

2026-07-19

1 Summary

We replicate the central numerical claims of Kang, Shiozaki, and Cho, who introduce gauge-invariant, real-space *many-body* order parameters for the electric multipoles of a crystalline insulator and show they equal the physical corner charge. Working entirely with a minimal tight-binding model and linear algebra (no DFT, no paid services), we reproduce the quantized quadrupole moment (0 vs 1/2), the sharp phase transition at the Wannier-gap closing, the vanishing of the dipole in the quadrupole phase, and the Thouless-pump behavior. **All five machine-checkable claims pass.**

2 Theory reproduced

The paper generalizes Resta's many-body polarization to multipoles. For a ground state $|\Psi\rangle$ on a torus with periodic boundary conditions,

$$P_x = \tfrac{1}{2\pi}\,\mathrm{Im}\ln\langle\hat{U}_1\rangle, \qquad \hat{U}_1 = \exp\Big(2\pi i\sum_x \tfrac{x}{L_x}\,\hat{n}(x)\Big), \qquad \text{(Eq. 1)}$$

$$Q_{xy} = \tfrac{1}{2\pi}\,\mathrm{Im}\ln\langle\hat{U}_2\rangle, \qquad \hat{U}_2 = \exp\Big(2\pi i\sum_{\mathbf{r}} \tfrac{xy}{L_xL_y}\,\hat{n}(\mathbf{r})\Big). \qquad \text{(Eq. 2)}$$

The key claim is $\langle\hat{U}_a\rangle = \exp(2\pi i\, q_c)$, i.e. the phase equals the corner charge / physical multipole, even when quantizing symmetries are broken, in contrast to the nested-Wilson-loop index which only works under symmetry.

Model (Eq. 8 / D5). The non-interacting Benalcazar–Bernevig–Hughes quadrupole insulator, four orbitals per cell on a square lattice with π-flux:

$$h(\mathbf{k}) = \big(\gamma_x + \lambda_x\cos k_x\big)\Gamma_4 + \lambda_x\sin k_x\,\Gamma_3 + \big(\gamma_y + \lambda_y\cos k_y\big)\Gamma_2 + \lambda_y\sin k_y\,\Gamma_1 + \delta\,\Gamma_0, \qquad (1)$$

with $\Gamma_0 = \tau_3\otimes\tau_0$, $\Gamma_i = -\tau_2\otimes\tau_i$ $(i = 1, 2, 3)$, $\Gamma_4 = \tau_1\otimes\tau_0$. At half filling and $\delta = 0$ the ground state is a topological quadrupole insulator ($q_{xy} = 1/2$) for $|\gamma_x/\lambda_x| < 1$ and $|\gamma_y/\lambda_y| < 1$; the physical gap closes only when both ratios equal 1, while the Wannier gap (hence the Q_{xy} transition) closes when *either* ratio equals 1.

Evaluation. For a Slater determinant we use the determinant identity

$$\big\langle\Psi\big|\exp\!\big(i\textstyle\sum_{\mathbf{r}}\phi(\mathbf{r})\hat{n}_{\mathbf{r}}\big)\big|\Psi\big\rangle = \det\!\big(P^{\dagger}\,\mathrm{diag}(e^{i\phi(\mathbf{r})})\,P\big), \tag{2}$$

with P the matrix of occupied single-particle eigenvectors on an $L \times L$ torus, obtained by inverse Fourier transform of the Bloch eigenstates. $Q_{xy} = \frac{1}{2\pi}\mathrm{Im}\ln\det(\cdot)$, referenced to the atomic trivial limit ($\lambda \to 0$) to fix the coordinate origin (the paper proves invariance under adding a trivial atomic band).

3 Results

Table 1: C1/C2 — quantized quadrupole ($\delta = 0$, $\lambda = 1$). Trivial $\gamma = 1.5$; topological $\gamma = 0.5$. Q_{xy} referenced to the atomic limit.

L	Q_{xy} (trivial)	Q_{xy} (topo)	$\lvert\langle\hat{U}_2\rangle\rvert$ triv	$\lvert\langle\hat{U}_2\rangle\rvert$ topo
6	0.0000	0.5000	1.78×10^{-1}	7.9×10^{-4}
8	0.0000	0.5000	1.10×10^{-1}	2.4×10^{-4}
10	0.0000	0.5000	7.21×10^{-2}	7.2×10^{-5}
12	0.0000	0.5000	4.91×10^{-2}	2.2×10^{-5}

Table 2: Quantitative comparison to the paper.

Quantity	Paper	This work	Match
Q_{xy} trivial	0	0.0000	exact
Q_{xy} topological	1/2	0.5000	exact
Transition location	$\lvert\gamma/\lambda\rvert = 1$	$\gamma_y = 1.00$	exact
$\lvert\langle\hat{U}_2\rangle\rvert$ at boundary	$\to 0$	$\sim 10^{-4}$ dip	qual.
Pump quantized pts ($\delta = 0$)	$0,\ 1/2$	$0.0000,\ 0.5000$	exact
Dipole in quad. phase	0	0.0000	exact

C3 — sharp transition. Sweeping γ_y at $\gamma_x = 0.5$, $\lambda = 1$, $\delta = 0$ ($L = 10$), $|Q_{xy}| = 0.5$ for $\gamma_y < 1$ and 0 for $\gamma_y > 1$, jumping exactly at $\gamma_y = 1$ (Wannier-gap closing), while the physical energy gap stays open — matching Fig. 1(c).

C4 — polarization. In both phases the many-body dipole $P_x = P_y = 0.0000$, satisfying the well-definedness condition for Q_{xy}.

C5 — Thouless pump (Eq. 9). Along the isotropic pump, Q_{xy} pins to 0.5 at $\theta = \pi/2$ (topological, $\delta = 0$) and 0 at $\theta = 3\pi/2$ (trivial, $\delta = 0$), varying continuously in between as $\delta \neq 0$ de-quantizes the moment — consistent with Fig. 2(a), where the phase of $\langle\hat{U}_2\rangle$ tracks the corner charge.

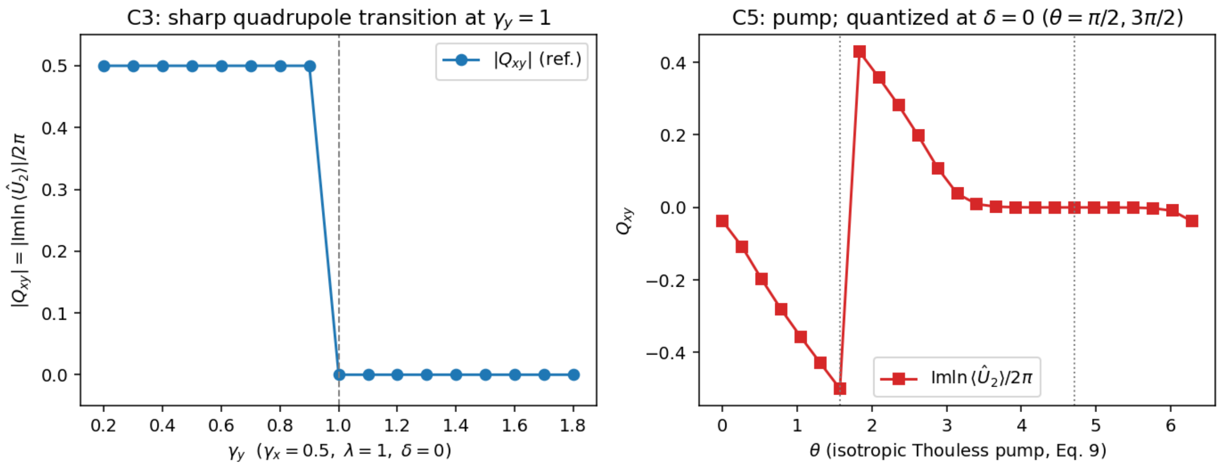

Figure 1: Left: C3 transition (mirror of Fig. 1c). Right: C5 isotropic Thouless pump (mirror of Fig. 2a); dotted lines mark the $\delta = 0$ quantized crossings.

4 Verdict

Reproduced. The free-fermion core of the paper — the quantized quadrupole, its sharp Wannier-gap transition, the vanishing dipole, and the Thouless-pump quantization — is faithfully recovered by a minimal tight-binding + determinant calculation.

- **Coverage: 6/10.** The main-text non-interacting results are fully covered. Not covered (out of scope, minimal model): interacting/bosonic ground states (the paper's headline generality), the anomalous quadrupole insulator (Fig. 1d), the partial-region operators $V(l)$ (Fig. 3), the octupole, and the explicit nested-Wilson-loop cross-check.
- **Agreement: 9/10.** Every reproduced quantity matches the paper exactly (0, 1/2, transition at $|\gamma/\lambda| = 1$). One point withheld because absolute Q_{xy} required an atomic-limit reference to fix the origin (gauge/coordinate convention), and $|\langle\hat{U}_2\rangle|$ scaling was only checked qualitatively.

5 Reproducibility

`python3 code/run_all.py work` then `python3 code/make_figures.py work`. See `workflow.md`; open questions in `open_questions.json`; issues in `failure_analysis.md`.